

MAT 192-003, Calculus and Analytic Geometry 2, Fall 2023
12:30 – 2:15 pm, Mondays and Wednesdays, Annex B, Room 114
Dr. Mary Kassebaum, mokassebaum@aacc.edu

Learn numerical and analytical techniques for integration and apply these techniques to solve problems involving definite integrals. Other topics include solving separable differentiable equations, extending the concepts of calculus to polar and parametric forms, using Taylor polynomials to approximate functions, and determining the convergence or divergence of improper integrals, infinite sequences, and infinite series.

Upon successful completion of this course, a student will be able to:

- a. Use definite integral to find the area, volume, work, arc-length, surface area, etc.
- b. Use advanced techniques of integration to evaluate definite and indefinite integrals.
- c. Solve separable differential equations, including applications.
- d. Approximate definite integrals numerically and analyze the error.
- e. Determine the convergence or divergence of improper integrals, infinite sequences and infinite series.
- f. Find Taylor Polynomials and apply them to finding approximations.
- g. Extend the concepts of calculus to curves in parametric and polar forms.

Homework Sessions with Dr. Kassebaum. Take the Homework Session Survey! See the full schedule on our Canvas site. Other times are available by request—please email to let me know when you can come.

Materials required for course:

- **WebAssign**—Please follow the instructions on our Canvas Website for MAT 192. You may enjoy free access for about 1 week, but then you will need to purchase access (one semester or multi-semester) if you do not already have it. An access code may be purchased online through the site or at the AACC bookstore. The full online textbook is included with this code. **Optional:** Paper copy of text: Calculus: Early Transcendentals, 9th ed., by James Stewart, published by Brooks/Cole, 2019.
- **Graphing calculator:** TI-83 or TI-84 is highly recommended. ***Portions of some exams and quizzes may be calculator-free, while other portions will require a graphing calculator.***
- **Scanning app or scanner** for scanning written work to pdf format to upload for grading.

Important facts, dates, and notices:

- **Exams** for this course will be during the first part of our regular class periods on **Wednesday, 9/27; Monday, 10/23; and Wednesday, 11/15**. The final exam is scheduled for **Wednesday, December 13, 12:30 – 2:30 pm**, in the regular classroom.
- Refund and audit deadlines are posted on MyAACC -> My Class Schedule. The last day to withdraw from this class is November 20, 2023.

Assignments, exams, and grading for this course:

- **WebAssign homework assignments are given on most class days.** The in-class question time for these assignments will be at the beginning of the next class period. In-class discussion will be limited to the most recent work, but please come to homework sessions or email me for help on ANY of the homework—I will be happy to help you. (WebAssign deadlines are posted in that software, with an automatically available 3-day extension.) WebAssign homework will count for 125 points of your final grade in the course, but extra credit can be earned for completing all assignments.
- The **Initial Skills Assessment**, 30 points, is due Wednesday, 8/30, in class or by 11:59 pm (submit online). The **Course Resource** assignment, 20 points, is due Wednesday, 9/6, in class or by 11:59 pm (submit online).
- There will be **four written “Graded Review” assignments to turn in** (or submit online) worth 30 points each. No class time is given to discussion of these assignments; however, you are encouraged to email me and/or attend homework sessions prior to the due date with any questions you may have. **Graded Review assignments turned in after the deadline are late**, and will be worth, at most, half credit for 48 hours after the deadline.
- There will be **ten note-taking assignments**, worth 3 points each. These must be submitted online by 12:30 pm on the days they are due. Extra credit is available on each assignment, so late submissions are not accepted.
- There will be **three 100-point exams** given during class time (note dates given previously). There will also be a **200-point comprehensive final exam** given on Wednesday, December 13, 12:30 – 2:30 pm. ***If you miss an exam, email me as soon as possible!***

- **11/30 – 12/6 is “MAT 192 – 003 makeup time.”** If you would like to retake **one** of the 3 exams of the course, you may do so 11/30 – 12/6, in the testing center (SUN 240). Instructions provided after exam 3.
- **Other assignments** may be given (including pop quizzes) if deemed necessary. Points earned will be included in the (dividing) point total of the course (**not** “extra credit.”)
- **Extra credit** assignments may be available periodically. They will always be announced in class or through CANVAS and will be open to all students in the course.
- **How grades will be assigned:** WebAssign homework (125 points); Graded Reviews (120 points); Initial Skills assessment (30 points); course resource assignment (20 points); Notes (30 points); exams (300 points); and the final exam (200 points) give a total of 825 points possible in the course (plus any extra graded assignments). You will receive an A for 90% or more of the total points, at least a B for 80% or more, at least a C for 70% or more, at least a D for 60% or more.

Dr. Kassebaum’s expectations of her students:

- **You are expected to attend class all the time, on time, and for the whole time.** If possible, please tell Dr. Kassebaum in advance if you cannot attend or must leave early. AACC requires a record of attendance, which becomes part of your academic record.
- **You are responsible for material covered in class whether you are present or not.** Either completed notes or a lecture video will be posted after each class period.
- **You are expected to use the CANVAS course site every week** to read your text with the study (note) guide, check course e-mail, check your grade, and access other course materials.
- **You are expected to use Canvas email to contact your professor (preferred)** or to send email to mokassebaum@aacc.edu. I try to respond to email within one business day.
- **You are expected to be courteous to your fellow students and to your professor.** Please be respectful in your speech and manner, as well as in your e-mail.
- **You are expected to turn off your phone** and all other electronic devices and stow them away prior to class. Failure to do this will result in a “knowledge demonstration opportunity” for the class.
- **Laptops, phones, or other electronic devices will not be used** for taking notes in class without advance permission from Dr. Kassebaum—*ASK FIRST!*
- This is a 4-credit course given in 15 weeks; **to be successful, you must spend at least 8-12 hours each week outside of class time studying and completing the work for this class.**
- **Do not wait to get help if you are confused!** USE my office times, e-mail me, USE the Math Tutoring Lab in the library (LIB 102), USE the Virtual Math Tutoring Center (through Canvas), USE our on-line classroom to create a study group.

Please understand...

I care about you, and I want you to be successful. I enjoy working with you, and I want you to see yourself as a mathematician and a future STEM professional.

If there is anything this semester—illness, work schedule, course policies, or assignment schedule—that is a barrier to your success, please contact me. It is my policy that “it never hurts to ask.” Please feel free to ask for what you need; if I can figure out how to provide it, I will.

The easiest way to ask is to send me an email through Canvas, or directly to mokassebaum@aacc.edu.

MAT 192-003 ESTIMATED Course Schedule

(Changes will be announced in class, through Canvas, or posted in WebAssign.)

Class meetings	In-class topics	Assignment deadline dates (see Canvas for times!)
Week 1: 8/28, 30	Brief review of Calculus 1 7.1, 7.2, 7.3	Notes 0 in Canvas by 12:30 pm, 8/30 Homework Session Survey, 8/30 Initial Skills Assessment, 8/31 WebAssign (WA) 7.1, 8/31
9/4: Labor Day	AACC closed 9/3-4	No classes
Week 2: 9/6	7.3, 7.4	Notes 1, due 9/6 Course Resource Assignment, 9/6 WA 7.2, 9/6
Week 3: 9/11, 13	7.4, 7.5 (light), 7.6 (light), 7.7	Notes 2, 9/11 WA 7.3, 9/11 WA 7.4, 9/13
Week 4: 9/18, 20	7.7, 7.8	Notes 3, 9/18 WA 7.5, 7.6, 9/18 WA 7.7 9/20
Week 5: 9/25, 27	7.8 EXAM 1, first part of class, 9/27 6.1	Graded Review for Exam 1, due 5 pm, Monday, 9/25 WA 7.8 9/25 Exam 1, 9/27
Week 6: 10/2, 4	Go over Exam 1 6.2, 6.3, 6.4	Notes 4, 10/2 WA 6.1 10/2 WA 6.2 10/4
Week 7: 10/9, 11	6.5, 8.1, 8.2, 9.1	Notes 5, 10/9 WA 6.3, 10/9 WA 6.4, 6.5, 10/11
Week 8: 10/16, 18	9.3, 9.4	Notes 6, 10/16 WA 8.1, 8.2, 9.1, 10/16 WA 9.3, 9.4, 10/18 Graded Review for Exam 2 due 5 pm, Friday, 10/20 (negotiable in class)
Week 9: 10/23, 25	Exam 2, first part of class 10/23 11.1 Go over Exam 2 11.2, 11.3	Exam 2, 10/23 Notes 7, 10/25 WA 11.1, 10/25
Week 10: 10/30, 11/1	11.3, 11.4, 11.5	Notes 8, 10/30 WA 11.2, 10/30 WA 11.3, 11/1
Week 11: 11/6, 8	11.6, 11.7	WA 11.4, 11/6 WA 11.5, 11.6, 11/8
Week 12: 11/13, 15	11.7, Review Exam 3, first part of class on 11/15 11.8	WA 11.7, 11/13 Graded Review for Exam 3, due 5 pm, Monday, 11/13
Week 13: 11/20, 22	11.8, 11.9, 11.10	Notes 9, 11/20 WA 11.8, 11/22
11/23 Thanksgiving	AACC closed 11/23 – 11/26	No classes
Week 14: 11/27, 29 Make-Up Exam 11/30 – 12/6	11.10, 11.11, 10.1, 10.2	Notes 10, 11/27 WA 11.9, 11/27 WA 11.10, 11/29
Week 15: 12/4, 6 Make-Up Exam 11/30 – 12/6	10.3, 10.4	Notes 11, 12/4 WA 11.11, 10.1, 12/4 WA 10.2, 10.3, 10.4, 12/6 Graded Review of Unit 4 due 5 pm, Friday, 12/8 (negotiable in class)
Finals Week: 12/11-15	Final Exam, Wednesday, 12/13, 12:30 – 2:30 pm, in classroom	Final Exam covers all material of course.

Emergency College Closure For closure information, students can check www.aacc.edu, and can sign up for text message alerts using the [Stay Informed with Campus Alerts](#) web page. If class is cancelled because of an emergency college closure, please check your course and email in Canvas for possible announcements.

Academic Integrity Policy Anne Arundel Community College, with a central mission of producing learning and a belief that individuals be given the opportunity to fully develop their potential, is committed to upholding rigorous and fair standards of student learning and achievement. Achieving successful student learning is dependent upon a dedication to academic integrity on the part of all members of the college community. Without academic integrity, students gain unfair advantage over others and impede their own development.

In support of this aim, Anne Arundel Community College requires all students to exhibit academic integrity in all their academic work. A culture of academic integrity, a unifying principle at AACC and all academic communities, is built upon respect for others' work, commitment to doing one's own work, and intolerance for academic dishonesty in all its forms. Students are responsible for reviewing and complying with the College's Academic Integrity Policy. A student found in violation of this policy shall be subject to appropriate sanctions, up to and including expulsion from the College, in accordance with the Resolving Allegations of Academic Misconduct Procedure and the Academic Misconduct Hearing Procedure. For more information, visit the [Academic Integrity Policy](#) and [AACC Policies webpage for the Academic Integrity Procedures](#) (.).

Student Conduct Policy Students shall always conduct themselves in a manner that demonstrates mutual respect and courtesy, displays appropriate standards of behavior, and refrains from any actions or in actions that impinge on the rights of others or disrupt the teaching and/or learning process or the operations of the college. A student found in violation of this policy or any other College policy shall be subject to appropriate sanctions, up to and including expulsion from the College, in accordance with the student conduct procedures. The full text of the policy is available on the college's [AACC Policies webpage](#) and in the Student Handbook and College catalog. Additional information for [Code of Conduct webpage](#).

Acceptable Use of Information Technology Resources Policy This policy governs the acceptable use of the college information resources by anyone. This policy applies to students enrolled in this course at any time they are using college resources. The goal of the usage policy is to encourage an environment of learning in which all students can interact in an open, legal, and ethical manner. The full text of the policy is available on the AACC [Acceptable Use of Technology Resources Policy webpage](#) that implement the policy are available on the AACC [Acceptable Use of Information Technology Resources Procedures](#).

Notice of Nondiscrimination updated June 1, 2023 The college prohibits all types of discrimination including, but not limited to, race – including hair texture and hairstyle – color, religion or creed, ancestry or national origin, sex, age, marital status, physical or mental disability, sexual orientation, gender identity, veteran status, citizenship, and genetic information. Complaints of discrimination shall be reported to the college's chief compliance and fair practices officer. For information regarding the college's compliance and complaints concerning discrimination, contact the chief compliance and fair practices officer and Title IX coordinator at complianceofficer@aacc.edu or 410-777-1239, or call Maryland Relay at 711.

Accommodations Students who need a reasonable accommodation should contact [Disability Support Services](#) at DSS@aacc.edu or 410-777-1411.

Religious Accommodations Policy Anne Arundel Community College prohibits discrimination, harassment, and retaliation on the basis of religion. The purpose of this Religious Accommodations Policy is to describe the rights of Employees and Students to request and receive Religious Accommodations in accordance with the College's policies and relevant laws and regulations regarding Religious Beliefs. For more information, review the [Religious Accommodations Policy webpage](#). Also, review the [Student Religious Accommodations Procedure webpage](#).

Free Support Services Available on Campus for AACC Students

Service	Location	Phone
Counseling Services	Student Union (SUN 203); appointment request via https://counseling.aacc.edu/TitaniumWeb/	410-777-7111 or email counseling@aacc.edu
Military/Veterans Resource Center	Truxal Library, room 112 https://www.aacc.edu/military/	410-777-4838 or email veterans@aacc.edu
Disability Support Services	Student Union (SUN 140), https://www.aacc.edu/resources/disability-support-services/	410-777-1411
Student Achievement and Success Program (SASP)	https://www.aacc.edu/about/administrative-offices/student-achievement-and-success-program/	410-777-2530
Substance Use Prevention, Education, and Resources	https://www.aacc.edu/resources/health-and-personal-counseling/substance-abuse-education-/#d.en.5787	410-777-2480 (Shawn Grim, RN, Health & Wellness Center)
Writing Center	Truxal Library, room 136, as well as virtual sessions https://www.aacc.edu/resources/academic-services/tutoring/writing-center/	Email jgrabin@aacc.edu (Dr. Jessica Rabin, coordinator)
Career Services	https://www.aacc.edu/resources/careers-and-employment/career-services/	Email jbsturtevant@aacc.edu
Student Engagement	Student Union (SUN 225) https://www.aacc.edu/about/administrative-offices/student-engagement/	410-777-2218
Student Food Pantry	Student Union (SUN 219) Glen Burnie Town Center (GBTC 311) Food pantry visits by appointment at AACC HelpLink appointment form	410-777-2218
HelpLink	Student Union (SUN 210) Glen Burnie Town Center (GBTC 311) https://www.aacc.edu/resources/health-and-personal-counseling/emergency-support/helpink-grant/	410-777-2511, or email studenthelpink@aacc.edu
AACC Rainbow Network	Look for Rainbow Network (ARN) signs around campus that identify offices of faculty and staff who work to end discrimination against individuals identifying as LGBTQ+ throughout our campus community and the community at large. https://www.aacc.edu/about/diversity-and-inclusivity/rainbow-network/	
Any student facing challenges securing their food, transportation, housing, or course materials is urged to contact the Office of Student Engagement for support. Call 410-777-2218. https://www.aacc.edu/about/administrative-offices/student-engagement/		